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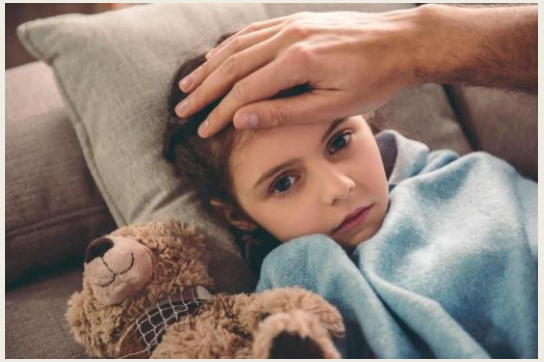
WINTER RESPIRATORY ILLNESSES



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OBJECTIVES

- Cover most common respiratory illnesses and how to treat them
  - Acute Bronchiolitis
  - Pertussis
  - Croup
  - Pneumonia
- How to identify and deffrenciate between similar conditions
- Most common causes of chest infections
- Complications of pneumonia

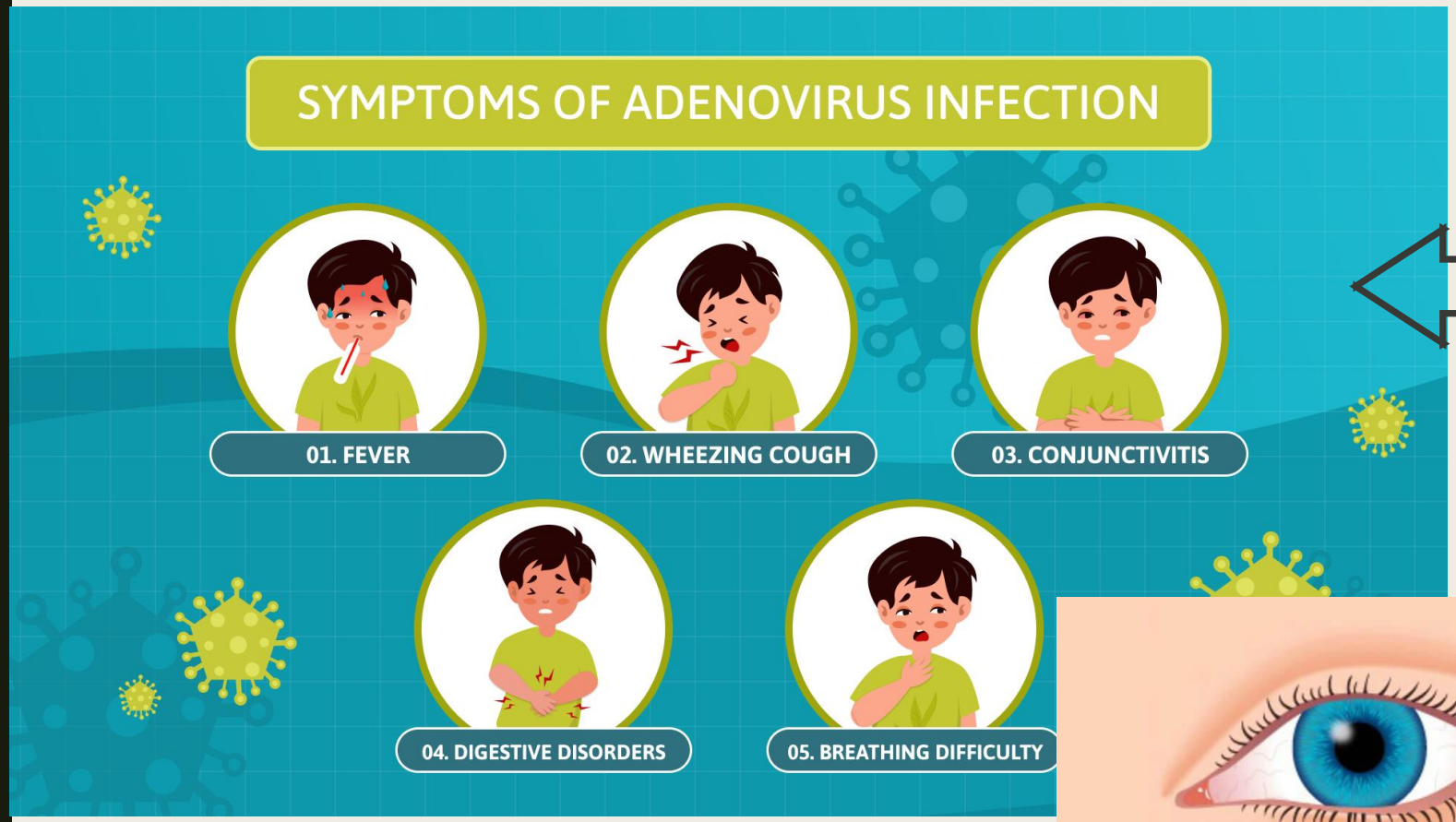
ACUTE BRONCHIOLITS

- Lower resp tract viral infection affecting <2yr olds.
- Peak incidence between 2-6 month of age.
- Characterized by URTI preceeding it.
- Typically presents with rhinorrhea, fever, cough, wheezes +/- creps, respiratory distress, +/- conjunctivitis.

- **Risk Factors:**
  - Prematurity <36 weeks
  - <3 months old
  - Airway anatomical abnormality
  - Immunodeficiency
  - Low birth weight
  - CLD
  - Congenital heart disease
  - Neurological disease

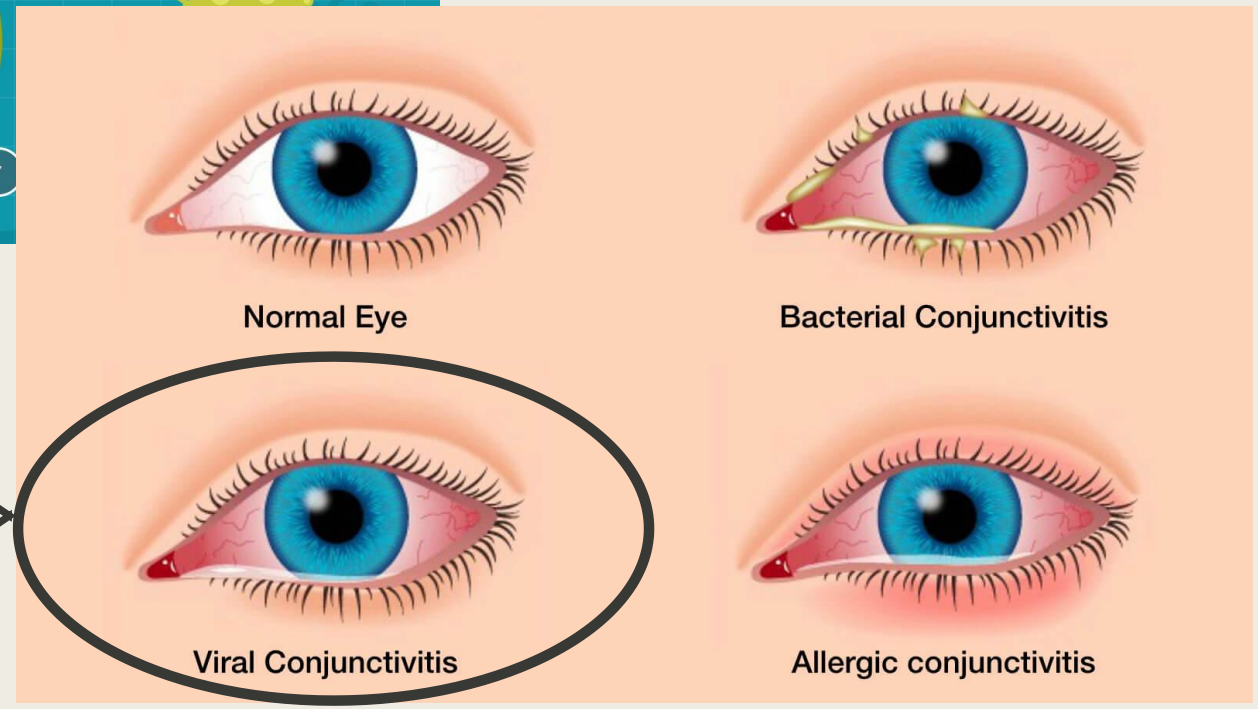
Viral causes in order of incidence

- **RSV** – peak in fall and winter.
- **Rhinovirus** – peak in spring and fall.
- **Parainfluenza virus** – type 3 peaks in spring and fall. Type 1&2 more common with croup.
- **Human metapneumovirus** – peaks in March and April, usually occurs with other viruses.
- **Influenza virus** – A and B subtype are treatable with Tamiflu
- **Adenovirus** – may also cause pharyngitis, more perulent secretions
- **Coronaviruses** – SARS-CoV-2, SARS-Cov-1, MERS-Cov.
- **Human bocavirus** – peaks in fall and winter, can cause pertussis-like illness.



Plenty of secretions!

Conjunctivitis with watery discharge. May be crusting.



Flu A & B have markedly elevated temperatures that partially respond to antipyretics. Generalized body aches, malaise, headache.

Clinical Features

- Fever, cough, resp distress following 1-3 days of URTI
- Self-limiting, most of unhospitalized children recover by 28 days.
- Peak of illness is day 3-5 then gradually resolve over weeks.
- Average length of hospitalization 2-5 days.
  Shortest in Rhinovirus, longest in RSV.
- Course is longer in those of high risk group.
- Wheezes resolve in 1 week. Cough takes 2-3 weeks to resolve.

COMPLICATIONS

- Mostly occurs to high risk group (apnea + resp failure)

- **Dehydration:** increased insensible losses, decreased intake, +/- vomitting.
- **Aspiration pneumonia**: cough and tachypnea with full stomach.
- **Apnea**: mostly in high risk group + RR <30 or >70, age <2months, on O_2.
  Risk factor to develop Resp Failure. No virology corelation.
- **Resp Failure**: More in high risk group and age <6weeks.
  Mucus plugs & atelectasis are common finding, resolve fast.
  Hypercapnic Resp Failure with fatigue require resp support.
- **Secondary bacterial infection**: uncommon 1.2%. Higher in those in PICU.
  Usually seen as abrupt worsening in fever and severity after several days of static course.

Bronchiolitis Obliterans

- Post-infectious BO usually (not exclusively) due to adenovirus infection.
- Results in diffuse lung disease (ILD)
- High resolution CT shows mosaic perfusion, air trapping, and bronchiectasis
- Predominantly affect gas exchange in the lung tissue
- Most need oxygen

DIAGNOSIS

- Made clinically.
- CXR and labs are only for severe, complicated, suspiciouns of another daignosis (hear failure, pneumonia, foreign body, etc).
- Nasopharyngeal swab to identify treatable causes Flu A + Flu B.
- Blood gas analysis when you suspect respiratory failure.

Management

- Mainly supportive: oxygenation + hydration
- **OUTPATIENT**: Adequate hydration
  Normal saline drops + suction
  Antipyretic
  +/- tamiflu
- **INPATIENT**: Hydration with IVF
  Normal saline drops + suction
  +/- tamiflu
  Bronchodilators if responds to trial
  3% saline neb → poor evidence, not routinely advised, may increase cough. Waranted trial in severe cases
  Glucocorticoids only in recurrent wheezy chest or family hx of asthma
  Respiratory support → HFNC, CPAP, M.V

PERTUSSIS

- A.K.A whooping cough or 100 day cough.
- Highly contagious
- Caused by bordetella pertussis
- Characterized by severe coughing spells that end with whooping sound with inhalation, and followed by vomitting.
- Highly fatal before the introduction of vaccine.
- Infants are fully vaccinated by the age of 6 months

STAGES

- **Catarrhal stage:** Similar to viral URTI. Low grade fever.
  Typically 1-2 weeks but shorter or absent in young infants.
  Cough gradually increases.

- **Paroxysmal stage:** Coughing spells increase in severity.
  Lasts 2-8 weeks
  Can include gagging, gasping, eye bulging.

- **Convalescent stage:** Cough subsides over weeks to months.

- Clinical diagnosis. Guided by NPS PCR or c/s. (result takes weeks)
- Non-specific x-ray findings.
- Lymphocytosis, leukocytosis.
- **Management includes**:
  -Hydration with least bothersome route
  -Avoid cough triggers: cold dry air, nasal suction, feeding.
  -Antibiotics: when given early (within 7 days) shortens the duration of symptoms and decreases transmission, and decreases severity.
- **Macrolides** are treatment of choice:
  → Azithromycin 10mg/kg OD x5 days
  → Clarithromycin 7.5mg/kg BD x 7 days (not recommended <1month old)
  → Erythromycin10mg/kg/6hr x 14 days (risk of pyloric stenosis)

COMPARISON

Acute Bronchiolitis	Pertussis
■ Continuious cough	■ Bouts of cough
■ Wheezes and creps at all times	■ Quiet chest between bouts
■ Cough improves after first week	■ Cough worsens after first week
■ Viral	■ Bacterial
■ Vomitting not related to cough if present	■ Post-tussive vomitting is cardinal
■ Fast recovery	■ Months to reach full recovery

CROUP

- Laryngotracheitis
- Inspiratory stridor, cough, hoarseness of voice.
- Characterized by “barking cough”
- Viral croup = Classic croup. Coryza and fever.
  6 months – 3 years
  self-limiting. Resolves in about 3 days.
- Spasmodic croup = Frequently recurrent croup
  6 months – 3 years
  self-limiting. Resolves fast. Recurrent
  usually runs in the family
- Viruses: RSV, influenza, parainfluenza, SARS-CoV-2, measles.

WESTLEY SCORE

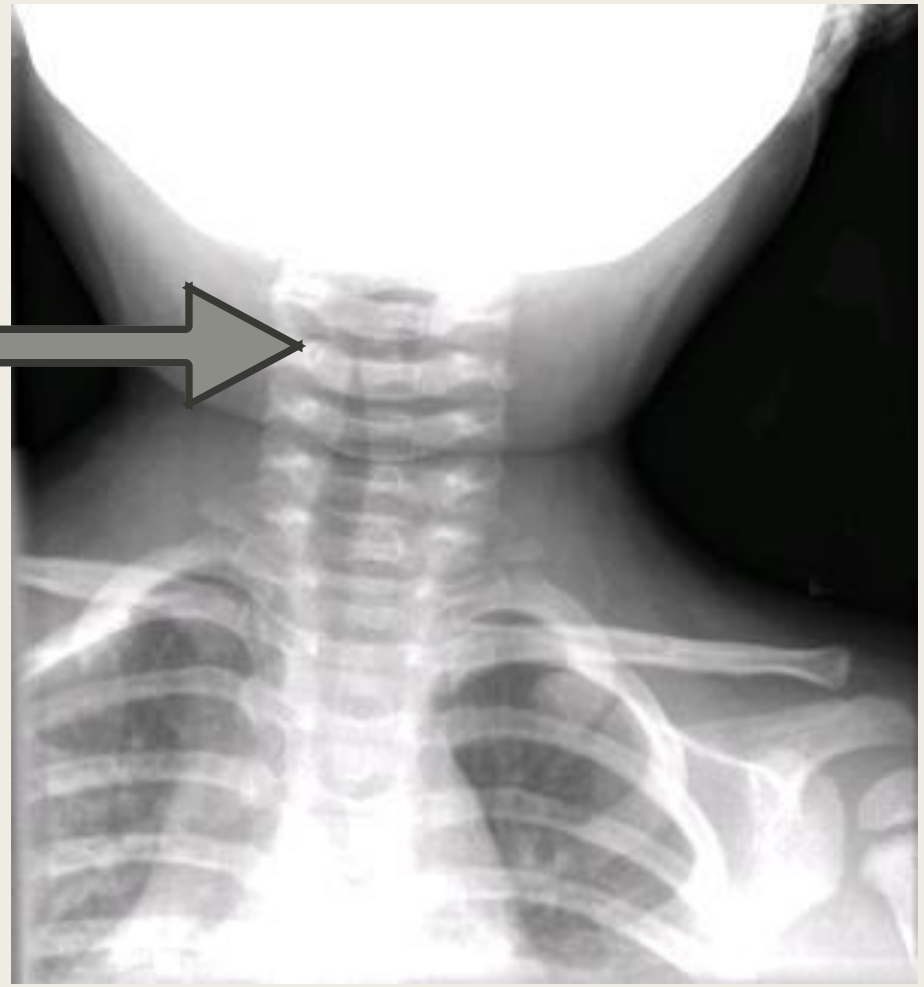
Clinical feature	Assigned score
Level of consciousness	
Normal, including sleep	0
Disoriented	5
Cyanosis	
None	0
With agitation	4
At rest	5
Stridor	
None	0
With agitation	1
At rest	2
Air entry	
Normal	0
Decreased	1
Markedly decreased	2
Retractions	
None	0
Mild	1
Moderate	2
Severe	3
Total	

Score total	Severity	Description	Management
≤2	Mild	Occasional barky cough, no stridor at rest, mild or no retractions	▪ Home treatment – Symptomatic care including antipyretics and oral fluids ▪ Outpatient treatment – Single dose of oral dexamethasone* 0.15 to 0.6 mg/kg (maximum 16 mg) or oral prednisolone (1 mg/kg)
3 to 7	Moderate	Frequent barky cough, stridor at rest, mild to moderate retractions	▪ Single dose of oral dexamethasone 0.6 mg/kg (maximum 16 mg)* ▪ Nebulized epinephrine¶ ▪ Hospitalization is generally not needed but may be warranted for persistent or worsening symptoms after treatment with glucocorticoid and nebulized epinephrine
8 to 11	Severe	Frequent barky cough, stridor at rest, marked retractions, significant distress	▪ Single dose of oral/IM/IV dexamethasone 0.6 mg/kg (maximum 16 mg)* ▪ Repeated doses of nebulized epinephrine¶ may be needed ▪ Inpatient admission is generally required unless marked improvement occurs after treatment with glucocorticoid and nebulized epinephrine
≥12	Impending respiratory failure	Depressed level of consciousness, stridor at rest, severe retractions, poor air entry, cyanosis or pallor	▪ Single dose of IM/IV dexamethasone 0.6 mg/kg (maximum 16 mg) ▪ Repeated doses of nebulized epinephrine¶ may be needed ▪ Intensive care unit admission is generally required ▪ Consultation with anesthesiologist or ENT surgeon may be warranted to arrange for intubation in a controlled setting

Diagnosis & Treatment

- Made clinically.
- Labs are of no value. Agitation → worsens croup
- Radiographs not required.
- Treat according to westly score.

STEEPLE SIGN



PNEUMONIA

- Lower resp tract infection affecting lung parenchyma
- Risk factors:
  low socioeconomic groups (overcrowding)
  CHD
  BPD
  CF
  Asthma
  SCD
  neuromuscular disorders
  GERD, TEF
  Immunodeficiency
- Causes can be viral or bacterial.

In cases of **aspiration** pneumonia, anaerobic oral flora is likely the cause:
- Anaerobic streptococci
- Fusobacterium spp
- Bacteroides spp

Common etiologic agents of pediatric pneumonia*

Microbial agent	Susceptible hosts
Bacteria	
Chlamydia trachomatis	First 3 months after birth
Mycoplasma hominis	First 3 months after birth
Treponema pallidum	First 3 months after birth
Ureaplasma urealyticum	First 3 months after birth
Staphylococcus aureus	Primarily children <5 years
Streptococcus pyogenes	Primarily children <5 years
Chlamydia pneumoniae	Primarily children ≥5 years
Mycoplasma pneumoniae	Primarily children ≥5 years
Streptococcus pneumoniae	All
Viruses	
Respiratory syncytial virus	Primarily children <5 years
Influenza A and B	Primarily children <5 years
Human metapneumovirus	Primarily children <5 years
Parainfluenza viruses 1, 2, and 3	Primarily children <5 years
Coronaviruses (229E C43, NL63, HKU1)	Primarily children <5 years
Adenoviruses	Primarily children <5 years
Rhinovirus	Primarily children <5 years

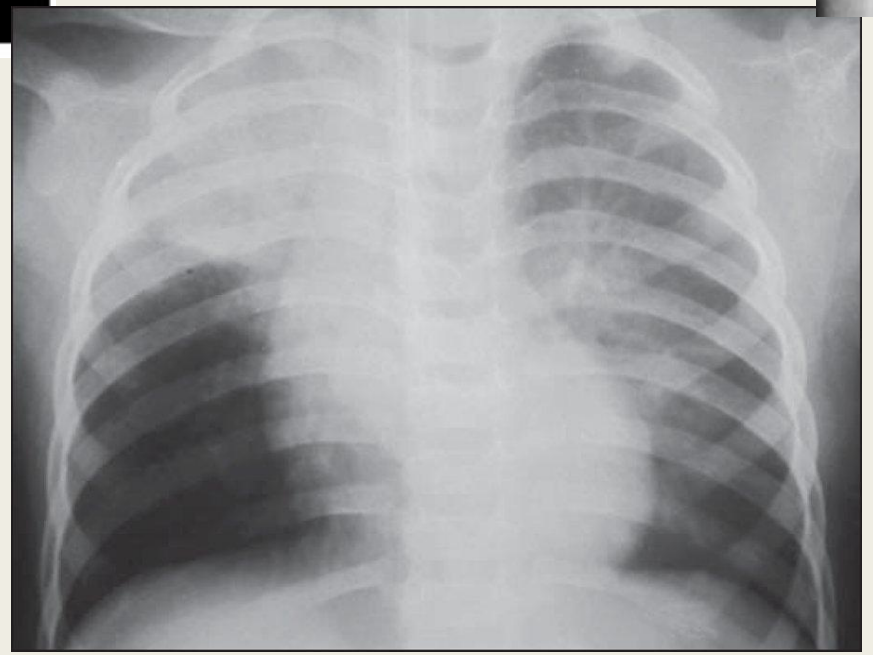
* Excluding neonatal pneumonia (in infants <28 days of age).

Presentation

- Combination of cough, fever, tachypnea.
- Tachypnea is the most accurate method to assess severity.
- Older children can complain of pleuritic chest pain.
- Occasionally abdominal pain with lower lobe pneumonia
- Atypical pneumonia (mycoplasma): dry cough, low fever, headache, malaise, myalgia, rash.
- Wheezing is likely to be viral pneumonia
- Crackles likely to be bacterial pneumonia
- Abrupt onset, high fever, ill appearing child → bacterial
- Localized findings → bacterial
- URTI, gradual progress, less severe → viral

Atypical pneumonia

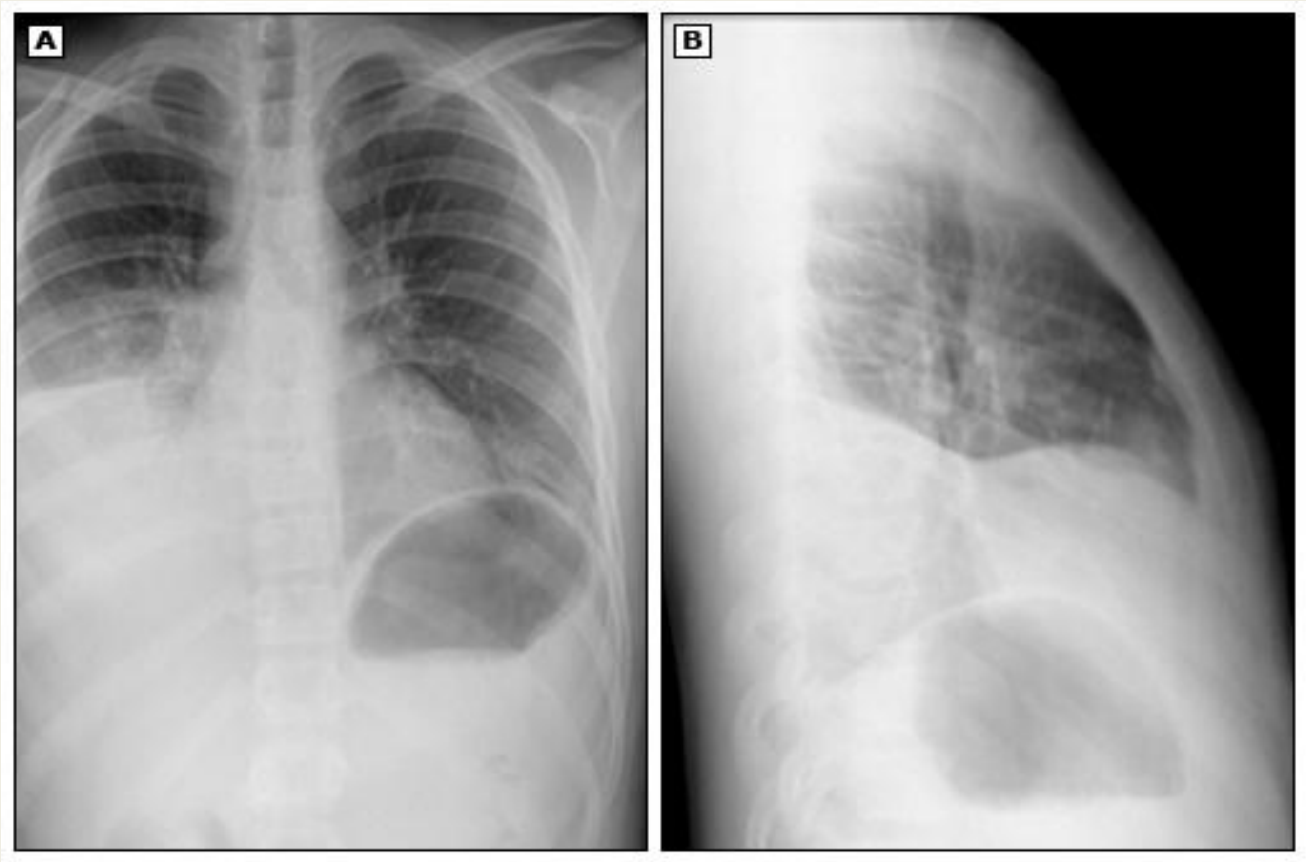
Aspiration pneumonia



Lower lobe pneumonia

Complications:

Pleural Effusion



- **Pleural tap:**
  Pus = empyema
  Turbid = inflammatory exudate
  Straw = transudate
  Bloody = malignancy, pulmonary infarction
  Milky white = chylothorax

Pleural Tap

- **Light's Criteria:** (1 +ve = exudate)
  -pleural protien/serum protein > 0.5
  -pleural LDH/serum LDH > 0.6
  -pleural LDH > 2/3 of lab's serum upper normal limit

- **Protein** > 4g/dL = TB effusion
  7 – 8g/dL = multiple myeloma

- **Cholesterol** > 45mg/dL = exudate

- **TG** > 110mg/dL = chylothorax

Pleural Tap

- **Glucose** < 3.3 mmol/L, pleural/serum glucose <0.5:

  -rheumatoid pleurisy
  -empyema
  -malignancy
  -TB
  -Lupus pleuritis
  -Esophageal rupture

- **Tumor Markers**
  CEA, CA125, CA 15-3, CA 19-9, CYFRA 21-1
- **Cultures**
- **Gram stain, Z-N stain**

Pleural Tap

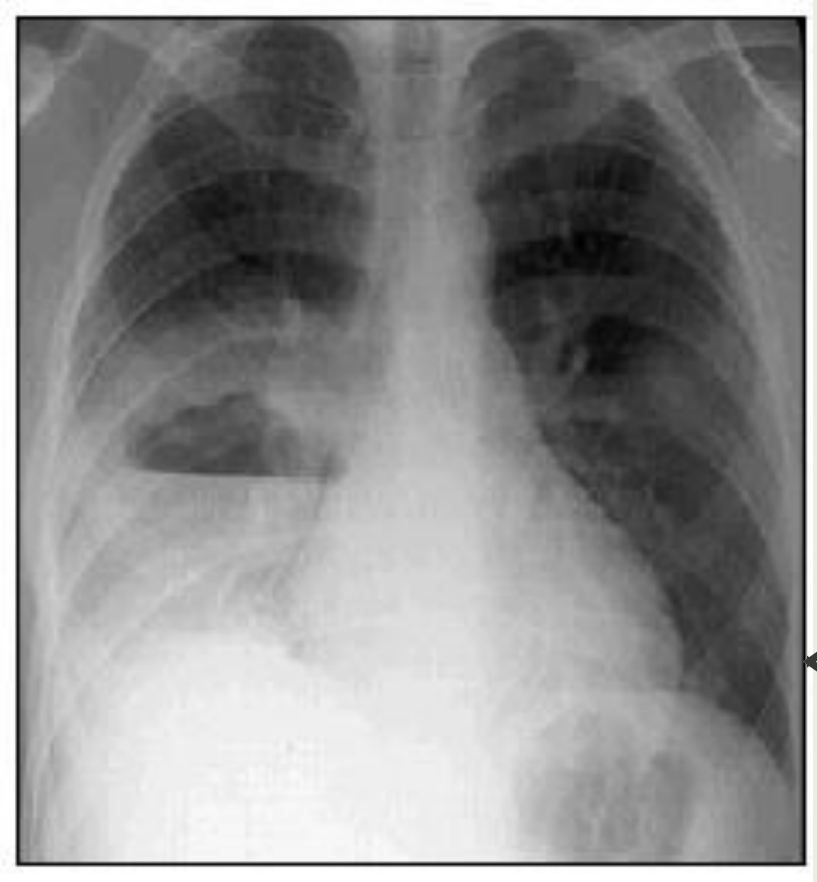
Transudate

- Heart Failure
- Nephrotic Syndrome
- Hypoalbuminemia
- SVC obstruction

Exudate

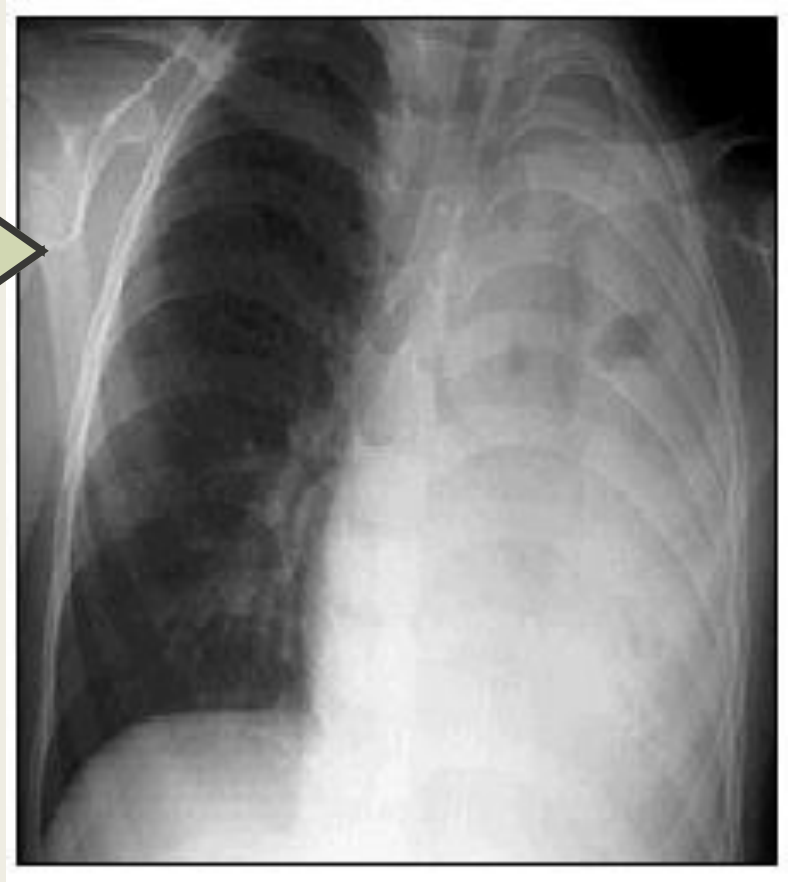
- Hemorrhage
- Infection
- Inflammation
- Malignancy
- Connective tissue dx
- Lymphatic

Complications



Necrotizing Pneumonia

Lung Abcess



Air fluid level.
Non-specific symptoms.
Lack of improvement.
Foul-smelling breath.

Sick looking, worsening of symptoms, septic, usually with pleural effusion.

Treatment (outpatient)

- Antipyretics & analgesics.
- Bronchodilators in case of wheezing
- Antibiotic choice based on age, and suspected pathogen

6 months to 5 years	
Typical bacterial*	Amoxicillin[¶] 90 mg/kg per day in 2 or 3 divided doses (MAX 4 g/day), **or**
	Amoxicillin-clavulanate 90 mg/kg per day of the amoxicillin component in 2 or 3 divided doses (MAX 4 g/day amoxicillin component)
	For children with mild reactions to a penicillin and no features of an IgE-mediated reaction[Δ]:
	▪ Amoxicillin 90 mg/kg per day in 2 or 3 divided doses (MAX 4 g/day), **or**
	▪ Amoxicillin-clavulanate 90 mg/kg per day of the amoxicillin component in 2 or 3 divided doses (MAX 4 g/day amoxicillin component), **or**
	▪ A third-generation cephalosporin, such as cefdinir 14 mg/kg per day in 2 divided doses (MAX 600 mg/day) in communities with a low rate of pneumococcal resistance to penicillin
	For children with IgE-mediated or serious delayed reaction to a penicillin:
	▪ Levofloxacin[◇] 16 to 20 mg/kg per day in 2 divided doses (MAX 750 mg/day), **or**
	▪ Clindamycin 30 to 40 mg/kg per day in 3 or 4 divided doses (MAX 1.8 g/day), **or**
	▪ Linezolid 30 mg/kg per day in 3 divided doses (MAX 1.8 g/day)
	In communities with a high rate of pneumococcal resistance to penicillin:
	▪ Levofloxacin[◇] 16 to 20 mg/kg per day in 2 divided doses (MAX 750 mg/day), **or**
	▪ Linezolid 30 mg/kg per day in 3 divided doses (MAX 1.8 g/day)

≥5 years	
Mycoplasma pneumoniae or *Chlamydia pneumoniae*	Azithromycin[¶] 10 mg/kg on day 1 followed by 5 mg/kg daily for 4 more days (MAX 500 mg on day 1 and 250 mg thereafter), **or**
	Clarithromycin 15 mg/kg per day in 2 divided doses (MAX 1 g/day), **or**
	Erythromycin 40 to 50 mg/kg per day in 4 divided doses (MAX 2 g/day as base, 3.2 g/day as ethylsuccinate), **or**
	Doxycycline 4 mg/kg per day in 2 divided doses (MAX 200 mg/day), **or**
	Levofloxacin[◇] 8 to 10 mg/kg once daily for children 5 to 16 years (MAX 500 mg/day); 500 mg once daily for children ≥16 years, **or**
	Moxifloxacin[◇ §] 400 mg once daily (≥18 years)
Typical bacterial*	Amoxicillin[¶] 90 mg/kg per day in 2 or 3 divided doses (MAX 4 g/day)
	For children with mild reactions to a penicillin and no features of an IgE-mediated reaction[Δ]:
	▪ Amoxicillin 90 mg/kg per day in 2 or 3 divided doses (MAX 4 g/day), **or**
	▪ A third-generation cephalosporin, such as cefdinir 14 mg/kg per day in 2 divided doses (MAX 600 mg/day)
	For children with IgE-mediated or serious delayed reaction to a penicillin:
	▪ Levofloxacin[◇] 8 to 10 mg/kg once daily for children 5 to 16 years (MAX 750 mg/day); 750 mg once daily for children ≥16 years, **or**
	▪ Clindamycin 30 to 40 mg/kg per day in 3 or 4 divided doses (MAX 1.8 g/day), **or**
	▪ Linezolid 30 mg/kg per day in 3 divided doses (MAX 1.8 g/day) for children <12 years; 20 mg/kg per day divided in 2 doses (MAX 1.2 g/day) for children ≥12 years
	In communities with a high rate of pneumococcal resistance to penicillin:
	▪ Levofloxacin[◇] 8 to 10 mg/kg once daily for children 5 to 16 years (MAX 750 mg/day); 750 mg once daily for children ≥16 years, **or**
	▪ Linezolid 30 mg/kg per day divided in 3 doses (MAX 1.8 g/day) for children <12 years; 20 mg/kg per day divided in 2 doses (MAX 1.2 g/day) for children ≥12 years

Aspiration pneumonia	
Community-acquired	Amoxicillin-clavulanate 40 to 50 mg/kg per day in 2 or 3 divided doses (MAX 1.75 g/day amoxicillin component)
	For children with mild reactions to a penicillin and no features of an IgE-mediated reaction[Δ]:
	▪ Amoxicillin-clavulanate 40 to 50 mg/kg per day in 2 or 3 divided doses (MAX 1.75 g/day amoxicillin component)
	For children with IgE-mediated or serious delayed reaction to amoxicillin:
	▪ Clindamycin 30 to 40 mg/kg per day divided in 3 or 4 doses (MAX 1.8 g/day)
	▪ Moxifloxacin[◊] 400 mg once daily (for ≥18 years)

Treatment (Inpatient):

- IVF to replace insensible losses
- Oxygen if required
- Antipyretics + analgesics
- Bronchodilator nebs if wheezes are present
- Respiratory support (HFNC, CPAP, MV)
- Antibiotics based on age and pathogen.
  Uncomplicated → 7 days total
  Complicated → 3-4 weeks total
- Tamiflu in cases of Flu A/B viral pneumonia

Age group and suspected pathogens	Suggested parenteral empiric agent(s)	Comments
1 to 6 months		
Bacterial (not *Chlamydia trachomatis* or *Staphylococcus aureus*)	One of the following: ▪ Ceftriaxone ▪ Cefotaxime	▪ **If CA-MRSA is suspected,** ADD one of the following: • Vancomycin or clindamycin • Ceftaroline* (alternative)
C. trachomatis	Azithromycin	
≥6 months		
Uncomplicated bacterial (not *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, or *S. aureus*)	One of the following: ▪ Ampicillin or penicillin G (preferred) ▪ Cefotaxime ▪ Ceftriaxone	▪ Cefotaxime and ceftriaxone are reserved for: • Children with incomplete Hib or *Streptococcus pneumoniae* immunizations, or • Communities with substantial prevalence of penicillin-resistant *S. pneumoniae* (eg, ≥25%)
M. pneumoniae or *C. pneumoniae*	One of the following: ▪ Azithromycin ▪ Erythromycin ▪ Levofloxacin	

Clinical syndrome (any age)	Suggested empiric parenteral agent(s)	Comments
Severe pneumonia	Combination therapy with one of the following: ▪ Ceftriaxone ▪ Cefotaxime **PLUS** one of the following: ▪ Azithromycin ▪ Erythromycin ▪ Doxycycline	▪ Children with severe infection may benefit from broad-spectrum therapy that addresses both typical and atypical pathogens ▪ **If *S. aureus* is a consideration, either**: • ADD vancomycin or clindamycin, OR • Provide therapy with ceftaroline* PLUS azithromycin
Severe pneumonia requiring ICU admission	Combination therapy with: ▪ Vancomycin **PLUS** one of the following: ▪ Ceftriaxone ▪ Cefotaxime **PLUS**: ▪ Azithromycin **PLUS**: ▪ Antiviral treatment for influenza if the child is hospitalized during influenza season	▪ **If *S. aureus* is likely**: • ADD nafcillin¶, OR • SUBSTITUTE linezolid for vancomycin and nafcillin, OR • Use ceftaroline* PLUS azithromycin PLUS antiviral treatment for influenza if the child is hospitalized during influenza season
Complicated pneumonia (eg, effusion/empyema, necrotizing process, abscess[Δ])	Combination therapy with one of the following: ▪ Ceftriaxone ▪ Cefotaxime **PLUS**: ▪ Clindamycin if *S. aureus* or anaerobic infection is a consideration	▪ Potential pathogens include *S. pneumoniae*, *S. aureus*, and *Streptococcus pyogenes* ▪ Vancomycin is an alternative to clindamycin for children with allergy to clindamycin or high prevalence of clindamycin resistance in the community[◇] ▪ Monotherapy with ceftaroline is an alternative if *S. aureus* is a consideration

What do they sound like?

Croup

Pertussis

Wheezing

Bronchiolitis

Take Home Messages

- Acute bronchiolitis is the most common infant illness during the winter. Requires hydration and oxygenation.
  Secondary bacterial infection is rare, avoid abx.
- Pertussis is characterized by intermittent bouts of cough and post-tussive vomitting.
- Croup: avoid agitation, treatment guided by Westley score.
- Pneumonia: RR is the best tool to assess distress.
- Treat pneumonia according to likely pathogen by age group.
- Complicated pneumonia → send to ER
- **Tusculman suppositories HAVE NO ROLE!** → could lead to CNS depression and are of no significant benefit.
- Cough syrup contraindicated in children under 12 years.

THANK YOU!

ANY QUESTIONS?